🐧 Linux Commands Cheat Sheet (with Examples & Explanations)

This document contains a **well-structured and detailed reference guide** for commonly used Linux commands. Each command includes **syntax, explanation, input, and output examples**.

🥧File & Directory Management

1. `pwd` – Print Working Directory

```
$ pwd
/home/prashant/Documents
```

Shows the current directory path.

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2. `ls` – List Directory Contents

```
$ ls
Documents  Downloads  file.txt  script.sh
```

Options:

- `ls -l` → Long format with details.
- `ls -a` → Show hidden files.
- `ls -lh` → Human-readable sizes.
- `ls -lt` → Sort by modification time.

```
$ ls -la
.  ..  .hiddenfile  file.txt  Documents
```

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3. `cd` – Change Directory

```
$ cd /var/log
$ pwd
/var/log

$ cd ..   # Move one level up
$ cd ~    # Go to home directory
$ cd -    # Switch to previous directory
```

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4. `mkdir` – Create Directory

```
$ mkdir projects
$ mkdir -p parent/child/grandchild
```

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5. `rmdir` – Remove Empty Directory

```
$ rmdir oldfolder
```

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6. `rm` – Remove Files or Directories

```
$ rm file.txt
$ rm -r folder   # remove directory recursively
```

⚠️Use with caution.

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7. `touch` – Create or Update Files

```
$ touch newfile.txt
$ touch file1 file2 file3   # multiple files
$ touch -c file.txt         # update timestamp if file exists
```

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8. `cat` – Display File Content

```
$ cat file.txt
Hello World

$ cat > notes.txt   # create new file
This is notes file.
<Ctrl+D>

$ cat file1 file2 > combined.txt
```

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9. `head` – Show First Lines

```
$ head file.txt
Line1
Line2
...
Line10

$ head -n 5 file.txt   # first 5 lines
```

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10. `tail` – Show Last Lines

```
$ tail file.txt
Line91
...
Line100

$ tail -n 5 file.txt
Line96
...
Line100

$ tail -f logfile.log   # follow logs in real-time
```

🦘Searching & Filtering

11. `grep` – Search Text Patterns

```
$ grep "error" logfile.log
Error at line 23
```

Common Options:

- `-i` → case insensitive
- `-n` → show line numbers
- `-v` → invert match
- `-c` → count matches
- `-r` → recursive search
- `-A/-B/-C n` → show context

```
$ grep -n "fail" app.log
45: login fail
```

---

12. `find` – Search Files

```
$ find /home -name "*.txt"
/home/user/file.txt

$ find /var -type f -size +10M
```

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13. `cut` – Extract Fields

File `emp`:

```
1 Rajat Manager 50000
2 Priya Developer 45000
```

```
$ cut -d " " -f2 emp
Rajat
Priya
```

---

14. `sort` – Sort File Lines

```
$ sort numbers.txt
1
2
3
...

$ sort -r numbers.txt   # reverse
$ sort -u numbers.txt   # unique
```

---

15. `uniq` – Remove Duplicates

```
$ sort sample.txt | uniq -c
3 apple
3 banana
3 orange
```

---

16. `sed` – Stream Editor

```
$ sed 's/old/new/g' file.txt
```

Replaces all `old` with `new`.

🐂File Info & Statistics

17. `wc` – Word, Line, Byte Count

```
$ wc myfile.txt
10 50 300 myfile.txt
```

- **10** → lines - **50** → words - **300** → bytes

```
$ wc -l myfile.txt   # lines
$ wc -w myfile.txt   # words
$ wc -c myfile.txt   # bytes
```

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18. `file` – File Type

```
$ file myfile.txt
myfile.txt: ASCII text
```

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19. `stat` – File Details

```
$ stat file.txt
Size: 1024  Blocks: 8  IO Block: 4096
Access: (0644/-rw-r--r--)
Modify: 2025-09-22 09:58:45
```

🐞Viewing Large Files

20. `more` – Page Viewer

```
$ more /var/log/syslog
```

Keys: - `space` → next page - `enter` → next line - `q` → quit

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21. `less` – Advanced Pager

```
$ less /var/log/syslog
```

Keys: - `f` → forward page - `b` → backward page - `/word` → search - `n/N` → next/previous match - `g/G` → start/end - `q` → quit

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🌍 Text Manipulation

22. `tr` – Translate Characters

```
$ echo "hello" | tr 'a-z' 'A-Z'
HELLO

$ echo "apple banana" | tr ' ' '\n'
apple
banana
```

- `-d` → delete characters
- `-s` → squeeze repeats

Example file:

```
This   is   jaaava
JAVA IS GREAT
```

```
$ tr -s 'a' < java
This   is   java
JAVA IS GREAT

$ tr -s ' ' < java
This is jaaava
JAVA IS GREAT
```

---

23. `paste` – Merge Files Line by Line

```
$ paste states capitals
Punjab Chandigarh
Haryana Chandigarh
Gujrat Ahmedabad
```

```
$ paste -d ":" states capitals
Punjab:Chandigarh
```

---

24. `echo` – Print Text

```
$ echo "Hello, World!"
Hello, World!

$ echo $USER
prashant

$ echo -e "Line1\nLine2"
Line1
Line2
```

Comparing Files

25. `diff` – Compare Line by Line

File1:

```
Hello
World
Linux
```

File2:

```
Hello
Word
Linux OS
```

```
$ diff file1 file2
2c2
< World
---
> Word
3c3
< Linux
---
> Linux OS
```

`2c2` means line 2 in file1 should be **changed** to match line 2 in file2.

---

26. `cmp` – Compare Byte by Byte

File1: `Hello` File2: `Hella`

```
$ cmp file1 file2
file1 file2 differ: byte 5, line 1
```

♎At byte 5, `o` vs `a`.

⛈️System Information & Monitoring

27. `date` – Current Date/Time

```
$ date
Mon Oct  1 10:15:30 UTC 2024
```

28. `cal` – Calendar

```
$ cal
 October 2024
Su Mo Tu We Th Fr Sa
       1  2  3  4  5
 6  7  8  9 10 11 12
```

29. `uname` – System Info

```
$ uname -a
Linux mypc 5.15.0-86-generic x86_64 GNU/Linux
```

30. `hostnamectl` – Host Info

```
$ hostnamectl
Static hostname: mymachine
Operating System: Ubuntu 22.04 LTS
Kernel: Linux 5.15.0
Architecture: x86-64
```

31. `who` – Logged-in Users

```
$ who
prashant  pts/0  2024-10-01 09:00
```

32. `uptime` – System Uptime

```
$ uptime
10:15:30 up 2 days,  3:15,  2 users,  load average: 0.23, 0.18, 0.11
```

33. `free` – Memory Usage

```
$ free -h
total   used   free
8.0G    3.2G   4.8G
```

34. `df` – Disk Usage

```
$ df -h
Filesystem  Size  Used  Avail  Use%  Mounted on
/dev/sda1   100G   50G    50G   50%  /
```

35. `du` – Directory Usage

```
$ du -h /var
4.0K /var/log
16K  /var/cache
```

🦝 Permissions & Ownership

36. `chmod` – Change Permissions

```
$ chmod 755 script.sh   # rwxr-xr-x
$ chmod u+x script.sh   # add execute for user
```

37. `chown` – Change Owner

```
$ chown user:group file.txt
```

Processes & System Control

38. `ps` – Show Processes

```
$ ps aux | head -5
USER   PID %CPU %MEM COMMAND
root     1  0.0  0.1 systemd
```

39. `top` – Process Monitor

Interactive real-time CPU/memory usage.

40. `htop` – Advanced Process Viewer

(better UI, needs installation).

41. `kill` – Terminate Process

```
$ kill 1234
```

42. `killall` – Kill by Name

```
$ killall firefox
```

🦏Archiving & Compression

43. `tar` – Archive Files

```
$ tar -cvf archive.tar files/
$ tar -xvf archive.tar
```

44. `gzip` / `gunzip`

```
$ gzip file.txt   # compress
$ gunzip file.txt.gz   # decompress
```

Networking

45. `ping` **– Check Connectivity**

```
$ ping google.com
```

46. `curl` **– Fetch Data**

```
$ curl https://example.com
```

47. `wget` **– Download Files**

```
$ wget https://example.com/file.zip
```

48. `ssh` **– Remote Login**

```
$ ssh user@host
```

49. `scp` **– Secure Copy**

```
$ scp file.txt user@host:/path/
```

50. `ifconfig` **/** `ip addr` **– Show IP**

```
$ ip addr
```

51. `netstat` **– Active Connections**

```
$ netstat -tulnp
```

🐔 Miscellaneous

52. `history` **– Show Command History**

```
$ history | tail -5
  501  ls
  502  cd /var
```

53. `man` – Manual Page

```
$ man ls
```

54. `clear` – Clear Screen

```
$ clear
```

55. `exit` – Logout

```
$ exit
```

✅This cheat sheet now has **commands → explanation → input/output examples** for easy revision and practical use.